

Example : $h_{a,b}(m) = a \cdot m + b \pmod{p}$

let p be a large prime ($1 < b < p$)

$$(a, b) \in \mathbb{Z}_p^2 = \mathcal{H}$$

$$m \in \mathbb{Z}_p = \mathcal{M} = \mathcal{C}_-$$

LEMMA Above $\mathcal{H} \rightsquigarrow$ PAIRWISE INDEPENDENT.

Proof. Fix any $m, m' \in \mathbb{Z}_p$, $m \neq m'$, $z, z' \in \mathbb{Z}_p$.

$$\Pr_{a,b} [h_{a,b}(m) = z \wedge h_{a,b}(m') = z']$$

$$P_{a,b} \left[\begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} z \\ z' \end{pmatrix} \right]$$

$$P_{a,b} \left[\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} m & 1 \\ m' & 1 \end{pmatrix}^{-1} \begin{pmatrix} z \\ z' \end{pmatrix} \right]$$

↳ The inverse exists
as $m \neq m'$

$$= \frac{1}{p^2}$$

~~1/1~~

THM Every PAIRWISE INDEPENDENT $\mathcal{H}$ defines an ε -stat. secure MAC with $\varepsilon = 1/|\mathcal{C}|$

$$\mathcal{H} = \{ h_K : \mathcal{M} \rightarrow \mathcal{C} \}$$

Proof. By Bayes, $\forall m, m', z, z',$
 $m \neq m'$

$$\Pr [z' = h_K(m') \mid z = h_K(m)] =$$

$$\frac{\Pr [\tau = h_K(m') \wedge \tau = h_K(m)]}{\Pr [h_K(m) = \tau]} =$$

$$\frac{1}{|\tau|^2} = \frac{1}{|\tau|}$$

Be! news:

THM

A my t -time 2^{-1} -secure
MAC has a key of size $(t+1) \cdot \lambda$.

RANDOMNESS EXTRACTION

How do Alice & Bob get random bits?

General to crypto:

- Keys must be uniform.
- Randomness also later needed for security (i.e. CPA Security).

You can think about this as a '2-step process':

1) Randomness extraction: Starting

from IMPERFECT source of randomness, generate "pure" RANDOMNESS.

2) Expand what you extracted as needed deterministically. (PSEUDORANDOMNESS).

3) Plus some way to refresh the RANDOMNESS that you can again expand.

Here is the question: We have a source X and we would like to design ext. function Ext s.t. $\text{Ext}(X)$ is UNIFORM.

Impossible in general (e.g. $X = \text{CONSTANT}$).

X should be UNPREDICTABLE to some extent.

DEF (MIN-ENTROPY) The MIN-ENTROPY of X

$$H_{\infty}(X) \doteq -\log \max_x \Pr[X=x]$$

Intuition: The more the MIN-ENTROPY the better. For example:

$$\text{Let } X = 0^M; H_{\infty}(X) = 0$$

Let $X \equiv U_m$ ($X \leftarrow \{0,1\}^m$)

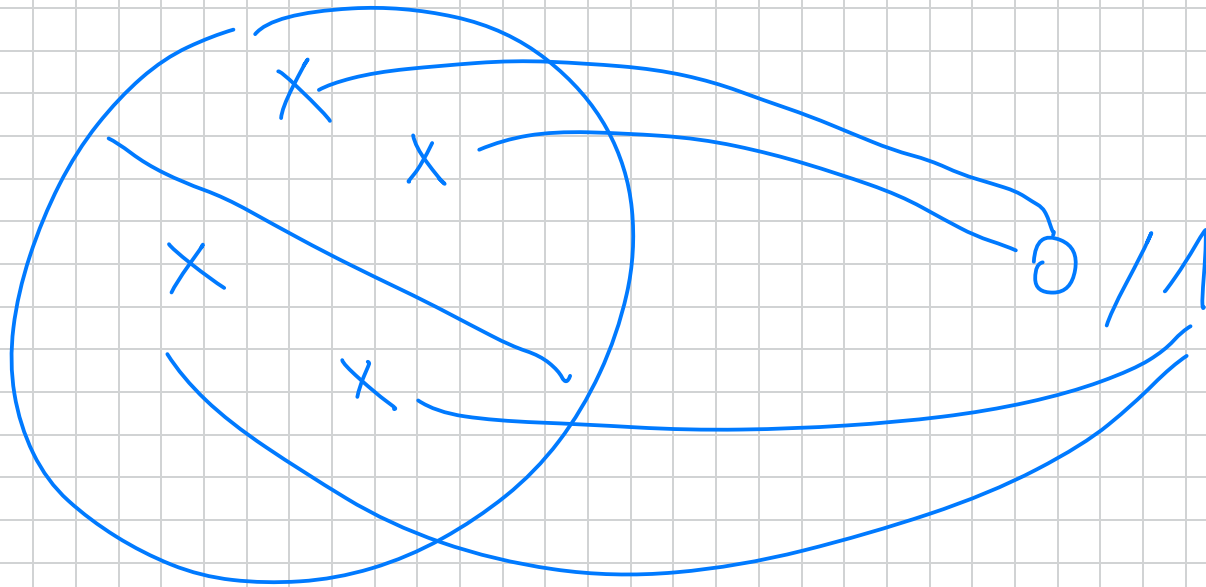
$$H_{\infty}(X) = m$$

Typically: One knows $H_{\infty}(X) \geq k$
for some $k < m$ and we'd like to
extract as much randomness as possible.

THM. There is no such $\text{Ext}: \{0,1\}^m \rightarrow \{0,1\}$
even if $k = m-1$.

Proof. Take any $\text{Ext}: \{0,1\}^m \rightarrow \{0,1\}$.

Let $b \in \{0, 1\}$ s.t. b maximize,
 $|Ext^{-1}(b)|$



$$\Rightarrow |Ext^{-1}(b)| \geq 2^{n-1}$$

Let X_{BAD} be UNIFORM over $Ext^{-1}(b)$.

$$H_{\infty}(X_{BAO}) \geq m-1 \quad \text{but}$$

$$\text{what is } \text{Ext}(X_{BAO}) = b \quad \text{constant!}$$

Jim

Take-away: No EXT can route all min-entropy sources. Approaches:

- Assume multiple independent sources,

2-SOURCE EXTRACTORS.

- Use crypto tools (e.g. HASH

FUNCTIONS). But requires strong assumptions on these functions.

— SEEDED EXTRACTION. We cover this.

Take: Family of extractions

$$\text{Ext} : \underbrace{\{0,1\}^{\ell}}_{\text{SEED}} \times \underbrace{\{0,1\}^M}_{\text{SOURCE}} \rightarrow \underbrace{\{0,1\}^l}_{\text{OUTPUT}}$$

The seed is going to be RANDOM.
Why not cheating?

- Seed as short catalyst; output
can be longer.

- Seed is going to be public
and RE USABLE.

The Adversary
knows it!

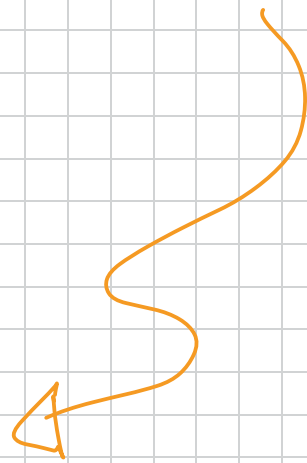
DEF Ext: $\{0,1\}^n \times \{0,1\}^m \rightarrow \{0,1\}^l$

is a (K, ϵ) ^{SEEDED} extractor if $\forall X$

s.t. $H_\infty(X) \geq K$ we have:

$$(S, \text{Ext}(S, X)) \stackrel{\sim}{\sim}_{\varepsilon} (S, U_{\ell}) (\equiv U_{d+\ell})$$

where $S \leftarrow U_d$



$\forall$ UNBOUNDED A can distinguish
 $(S, \text{Ext}(S, X))$ from $U_{d+\ell}$

w.p. $\leq \varepsilon$.

$$X \approx_{\varepsilon} Y \Leftrightarrow SD(X; Y) \leq \varepsilon$$

$$SD(X; Y) = \frac{1}{2} \sum_z |P_X[X=z] + P_Y[Y=z] - P_{X,Y}[X=z, Y=z]|$$

THM (LEFT-OVER HASH LEMMA) Let

$$\mathcal{H} = \{h_s : \{0,1\}^n \rightarrow \{0,1\}^l \mid s \in \{0,1\}^o\}$$

be PAIRWISE INDEPENDENT.

Then $\text{Ext}(s, x) = h_s(x)$ vs

a (k, ϵ) -seeded extractor for

$$k \geq l + \underbrace{2 \log(1/\epsilon) - 2}$$

MIN-ENTROPY

GAP

The proof will use the following
lemma:

LEMMA Let Y be a RV over $\mathcal{Y}$.

Assume:

$$\text{Col}(Y) = \sum_{y \in \mathcal{Y}} p_Y[Y=y]^2$$

$$\begin{array}{l} \text{Uniform} \\ \text{over } \mathcal{Y} \end{array} \leq \frac{1}{|\mathcal{Y}|} (1 + 4\varepsilon^2)$$

$$\Rightarrow \text{SD}(Y; U) \leq \varepsilon.$$

Proof (THM). Set $\gamma = (S, h(S, X))$

$$\text{Col}(\gamma) = \Pr[\gamma = \gamma']$$

$$= \Pr[S = S' \wedge h(S, X) = h(S', X')]$$

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$$= \underbrace{\Pr[S = S']}_{\text{Col}(S)} \cdot \Pr[h(S, X) = h(S, X')]$$

$$\begin{aligned}
 &= 2^{-o} \cdot \left(\Pr[X = X'] \cdot 1 + \Pr[h(S, X) = h(S, X') \wedge X \neq X'] \right) \\
 &\quad \swarrow \\
 &H_{\infty}(X) \geq k
 \end{aligned}$$

$$\leq 2^{-o} \left(2^{-k} + \Pr[h(S, X) = h(S, X') \wedge X \neq X'] \right)$$

$$\leq 2^{-o} \left(2^{-k} + 2^{-l} \right)$$

$$= \frac{1}{2^{d+l}} \left(2^{l-k} + 1 \right)$$

$$l-k \leq 2 - 2\log(1/\varepsilon)$$

$$\leq \frac{1}{|Y|} \left(2^{2 - 2\log(1/\varepsilon)} + 1 \right)$$

$$= \frac{1}{|Y|} \left(1 + 4\varepsilon^2 \right)$$

$$\Rightarrow SD(Y; U) \leq \varepsilon \quad \square$$